

Min Max Seo (They/He)

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EDUCATION

Ontario Tech University, ON, Canada

Master of Arts – Education (MA Thesis in Mathematics Education)

Sep 2024 – Aug 2027 (Expected)

Thesis Supervisor: Dr. Ami Mamolo

University of Waterloo, ON, Canada

Bachelor of Arts – Honours Psychology, Joint Honours Mathematics

Sep 2020 – Aug 2024

AWARDS, HONOURS & GRANTS

Samuel and Sonya Na Scholarship, Korean Canadian Scholarship Foundation (\$5,000)	2024
Kerr Ostrander Leadership for Tomorrow Award, University of Waterloo (\$2,500)	2024
Sameer Ahuja Perseverance Award, University of Waterloo (\$3,000)	2024
Weston Family Scholarship, Weston Family Foundation (\$37,400)	2021-2024
WMCTF Mathematics Teaching Fellowship, University of Waterloo (\$500)	2023
Piquard Family Award, University of Waterloo (\$1,200)	2022
Kay and Stuart Nicol Scholarship, University of Waterloo (\$2,500)	2022
University of Waterloo Merit Scholarship, University of Waterloo (\$1,000)	2021

RESEARCH INTERESTS

I am a trans/nonbinary mathematics education researcher with a lived experience of homelessness. With my various lived/living experiences, my research focuses on math education, particularly on teacher education and the math education of marginalized groups. Additionally, my research interests also include youth homelessness and substance use.

- Designing Accessible K-12 Mathematics Curriculum
- Teacher Education: Transferable Impressions of Mathematics
- Supporting Students with Various Learning Needs in Mathematics Education
- Mathematics Education for Neurodivergent Students
- Youth Homelessness and Substance Use

PRESENTATIONS & TALKS

Kirst, M., **Seo, M.**, Sykes, J., Rozycki, V., Delair, L., Iuliani, L., Laing, S., Graham, J. (2025, February 24). *Examining the effectiveness of integrated housing, mental health, and addiction service models for youth experiencing homelessness* [Conference Presentation]. The International Conference on the Prevention of Youth Homelessness, Toronto, ON, Canada.

Bates, S., **Seo, M.**, Harris, J. (2024, October 29). *Healthy and housed* [Conference Presentation]. CAEH24: National Conference on Ending Homelessness, Ottawa, ON, Canada.

Seo, M., Feng, A. (2024, June 2). *Presenting an emerging project about learning and teaching of mathematics: supporting current and future teachers of Ontario* [Gallery Walk]. Canadian Mathematics Education Study Group Annual Conference, Université Laval.

Seo, M., McDiarmid, S., Henderson, H. (2024, April 10). *How to assess children's sense of belonging in non-school contexts: A validation study of children's community and program-level belonging survey for ages 8 to 13* [Poster Presentation]. Psychology Discovery Conference, University of Waterloo.

Seo, M. (2024, February 27). *How to become an ally for colleagues and participants in the 2SLGBTQ+ community* [Presentation]. Safer Use Program, House of Friendship.

RESEARCH EXPERIENCE

Faculty of Education, Ontario Tech University – MA Student Supervisor: Dr. Ami Mamolo (Mathematics Education)	Sep 2024 – Present
Department of Psychology, Wilfrid Laurier University – Collaborating Researcher PI: Dr. Maritt Kirst (Community Psychology)	Jan 2023 – Present
Social Development Lab (SDL), University of Waterloo – Research Assistant PI: Dr. Heather Henderson (Developmental Psychology)	Dec 2023 – Aug 2024
Faculty of Education, Wilfrid Laurier University – Research Assistant PI: Dr. Jennifer Holm (Mathematics Education)	Jun 2022 – Aug 2024
Reasoning and Decision Making Lab, University of Waterloo – Research Assistant PI: Dr. Jonathan Fugelsang (Cognitive Psychology)	Jun 2022 – Apr 2024

PROJECTS

Canadian Housing First Adaptation Research Initiative (CHARI) – Collaborating Researcher <u>PI: Dr. Maritt Kirst, WLU</u> The goal of CHARI is to develop and strengthen partnerships through which to identify promising Housing First (HF) adaptations and conduct research on the needs of specific populations experiencing homelessness, in order to advance implementation of innovative, evidence-based HF programming in Canada. - Submitted a grant application to the SSHRC as a collaborator - Attend meetings regularly	May 2026 – Present
Master's Thesis in Mathematics Education – MA Student <u>Advisor: Dr. Ami Mamolo, OTU</u> By analyzing scripts written by pre-service teachers about supporting students' debugging processes, this project aims to explore how pre-service teachers encourage or hinder the construction of a conjecturing atmosphere. - Conducted a literature review - Analyze scripts written by pre-service teachers and organize them into themes	Aug 2025 – Present
Restart Project – Youth Advisor, Collaborating Researcher <u>PI: Dr. Maritt Kirst, WLU</u> Restart Project supports Housing First for Youth (HF4Y) by contributing to the knowledge of its long-term impact as a solution to youth homelessness in Canada. - Participated in monthly virtual meetings to advise on research project designs - Partnered on the grant proposal for the project - Participated in training to conduct qualitative research - Presented the qualitative results at conferences - Reviewed manuscripts for publication	Jan 2023 – Aug 2025
Validation of Waterloo Region Belonging Survey – Early Childhood Ver. – Research Assistant <u>PI: Dr. Heather Henderson, UW</u> The Waterloo Region Belonging Survey – Early Childhood Version (WRBS-Early) is designed to measure children's (ages 3–7) sense of belonging in their community programs. Cleaned and organized the data collected using R	May 2024 – Aug 2024
Learning and Teaching of Mathematics – Coordinator Collaborated with other coordinators to create an online platform for learning and teaching of K-12 mathematics to reduce students' learning barriers and equip teachers to better guide students with additional learning challenges. Recruited other coordinators to lead the project	Nov 2023 – Aug 2024

- Subdivided the project into five different initiatives
- Organized bi-weekly meetings

Validation of WRBS-Middle – Directed Studies Student

Dec 2023 – Apr 2024

PI: Dr. Heather Henderson, UW

The Waterloo Region Belonging Survey – Middle Childhood Version (WRBS-Middle) is designed to measure children’s (ages 8–13) sense of belonging in non-school contexts.

- Cleaned and organized the quantitative data using SPSS
- Examined the internal consistency, validity, and factor structure of the WRBS-Middle using SPSS
- Wrote a formal report on the validation processes and results

Bet on Yourself (BOY) – Research Assistant

Jan 2023 – Apr 2024

PI: Dr. Jonathan Fugelsang, UW

BOY is an in-person study that investigates how people process information. A series of logical problems were presented for each participant to solve, and they were asked to “bet on themselves” by rating their confidence level on each response.

- Recruited and provided participation credits to participants on SONA (psychology research experience online platform)
- Oriented research participants to the study setting for data collection
- Participated in the online session of data cleaning and data analysis using R
- Proofread and reviewed the study before running the experiment

Thinking in Action (TIA) – Research Assistant

Jan 2023 – Nov 2023

PI: Dr. Jonathan Fugelsang, UW

TI was an in-person study investigating how people process information when reasoning and making decisions.

- Recruited and provided participation credits to participants on SONA
- Oriented research participants to the study setting for data collection
- Proofread the study information

Cueing Confidently (CC) – Research Assistant

Jan 2023 – Nov 2023

PI: Dr. Jonathan Fugelsang, UW

CC is an in-person study examining how people think and reason.

- Participated in SONA researcher training
- Oriented research participants to the study setting for data collection
- Recruited and provided participation credits to participants on SONA

Teachers’ Use of Mathematics Apps – Research Assistant

Sep 2022 – Mar 2023

PI: Dr. Jennifer Holm, WLU

This qualitative study aimed to examine teachers’ experience using math apps, mainly *DreamBox* and *Prodigy*, in their classrooms.

- Conducted a literature search and wrote a report of the results
- Participated in the research design meetings

TEACHING INTERESTS

- Mathematics education
- Mathematics: K-12 math, calculus, linear algebra, elementary number theory, mathematical logic
- Psychology: introductory psychology, educational psychology
- Education: special education, educational technology
- Research Methods: basic research methods and analysis, introductory statistics, programming in R

TEACHING EXPERIENCE

Mathematics Tutor

Oct 2017 – Present

Provided tutoring for curriculum-based and competition mathematics for students ranging from Grade 7 to first-year university, both in-person and online.

- Offer free/pay-what-you-can tutoring for students with additional financial and learning challenges
- Create an initial assessment using LaTeX for the group placements and curriculum development
- Design lesson materials based on various provincial curricula in Canada

Ontario Tech University – Teaching Assistant

Sep 2024 – Apr 2026

Actively supported students and graded students' work as a teaching assistant for the following courses:

- EDST 4470U – Trends and Issues in Educational Technology (Spring/Summer 2026)
 - 100 students enrolled
 - 105 hours assigned
- EDST 2700U – Special Education Foundations (Spring/Summer 2026)
 - 93 students enrolled
 - 35 hours assigned
- EDST 3710U – Diversity of Educational Needs (Winter 2025)
 - 56 students enrolled
 - 70 hours assigned
- EDST 3999U – Special Topic: Academic Skills for Education (Winter 2025)
 - 80 students enrolled
 - 70 hours assigned
- EDST 1000U – Foundations of Learning (Fall 2024, Fall 2025)
 - 300+ students enrolled
 - 140 hours assigned for each semester

Centre for Education in Mathematics and Computing – Math Circles Assistant

Sep 2022 – Dec 2022

The Centre for Education in Mathematics and Computing (CEMC) is housed within the Faculty of Mathematics at the University of Waterloo.

- Designed, assisted, and delivered CEMC Math Circles in-person and online for Grades 6–8 students to enrich their skills and interests in mathematics
- Supported the organization of Grades 9–12 CEMC Math Circles
- Proofread the Beaver Computing Challenge
- Participated in the Canadian Senior and Intermediate Contests marking sessions

TECHNICAL SKILLS

- Programming Languages: C, Python, Racket, Scheme, R, LaTeX, Turing, Java
- Languages: English, Korean
- Data Analysis Tools: SPSS, RStudio

PROFESSIONAL EXPERIENCE

Safer Use Program, House of Friendship – Harm Reduction Worker

Nov 2023 – Jul 2026

Safer Use Program (SUP) is an urgent medical care site within a local shelter in Waterloo, ON

- Encourage safer use practices and share harm reduction information with participants
- Facilitate a judgment-free environment by practicing trauma-informed care
- Provide interventions such as injectable naloxone and oxygen during incidents of accidental drug poisoning

Global Affairs Canada – Technical Advisor, Data Innovation

Nov 2021 – Dec 2021

Global Affairs Canada (GAC) manages Canada's diplomatic relations, promotes international trade and provides consular support.

- Moved GAC's data action plan forward by developing a SharePoint naming convention and a thematic list
- Participated in daily meetings on MS Teams

LEADERSHIP & COMMUNITY INVOLVEMENT

The Centre for Addiction and Mental Health – NYAC Member**Jun 2021 – Present**

National Youth Action Council (NYAC) comprises a community of passionate youth mental health advocates across Canada.

- Participate in research, service planning, and education opportunities at CAMH
- Facilitate an affirming environment to talk about mental health and substance use challenges

Measurement-Based Care Advisory Committee – Youth Advisor**Dec 2023 – Sep 2024**

MBC Youth Advisory Committee advises on Measurement-Based Care (MBC) at the Centre for Addiction and Mental Health and related Youth Hubs.

- Developed the curriculum videos for service providers as an actor
- Participated in monthly virtual meetings to advise services

University of Waterloo Orchestra – Violist**Jan 2024 – Apr 2024**

Participated in weekly rehearsals to perform in an end-of-term concert.

- Beethoven Symphony No. 6
- Grieg: Piano Concerto in A minor
- Weber, Overture to Der Freischütz

LIFT Youth Experts Collective – Youth Advisor**Sep 2021 – Nov 2023**

LIFT YEC is a youth committee to advise on the Individual Placement and Support (IPS) program at the Centre for Addiction and Mental Health.

- Participated in monthly virtual meetings to advise services and research project designs
- Drafted youth-friendly materials to explain the research project on the IPS program
- Advised on the materials to train staff members of youth hubs across Canada on the education/employment support program

The Glow Centre for Sexual and Gender Diversity – Trans Equity Director**Jan 2023 – Aug 2023**

The Glow Centre is the oldest-running university-based 2SLGBTQ+ group in Canada.

- Ensured that trans* students' voices were heard
- Advocated for the resolutions to the issues trans* students at the University of Waterloo were facing
- Maintained the Trans Resource Library

Waterloo Ready Mentorship, Math Orientation – Peer Mentor, Leader**Jun 2021 – Aug 2021**

The Waterloo Ready Mentorship program supports first-year students in transitioning smoothly into their first year at the University of Waterloo.

- Facilitated weekly hangouts for 30 first-year math students

The Women's Centre, University of Waterloo – Volunteer**Sep 2020 – Apr 2021**

The Women's Centre aims to provide a positive and supportive environment on campus for all women and trans* people.

- Received training to provide peer support for marginalized individuals
- Organized and supported events
- Provided feedback on trans* inclusion

Youth Committee, Friends of Ruby – Committee Member**Feb 2020 – Jan 2021**

Friends of Ruby is an organization dedicated to the progressive well-being of 2SLGBTQ+ youth (ages 16-29) through mental health, social, and housing services.

- Attended staff training and meetings to obtain general information about the organization
- Organized bi-weekly meetings to develop and manage programs for 2SLGBTQ+ youth
- Participated in discussions with the head directors of the organization
- Communicated with youth to hear their needs
- Documented outcomes of projects and future suggestions in an annual report

SELECTED CERTIFICATIONS

- Certification of Completion – Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans Course on Research Ethics (TCPS 2: CORE)
- Certification – Standard First Aid with CPR-C
- Certification of Completion – Crisis Intervention Training
- Certification of Completion – Applied Suicide Intervention Skills Training (ASIST)
- Certification – Trauma-Informed Care (TIC)
- Certification – Personal Health Information Protection Act (PHIPA) Training

SELECTED PROFESSIONAL DEVELOPMENT

- Canadian Mathematics Education Study Group Annual Meetings (2022, 2024, 2025, 2026)
- Building Capacity in the Scholarship of Mathematics Education (2025)
- Making the Shift: International Conference on the Prevention of Youth Homelessness (2025)
- Fields MathEd Forum (2024-2025, 2025-2026)
- CMS MathEd Online Meeting (2024, 2025)
- In and Of the World: Framing Mathematics (2024)
- Supporting Teachers' Mathematical Thinking for Society (2024)
- Mathematics and Mathematical Thinking for Society (2024)
- CAEH24: National Conference on Ending Homelessness (2024)
- #Frayme2023
- Trans Math Day (2022)
- Gender Equity in the Mathematical Study (GEMS) of Combinatorics (2021)
- Mathematical Summer in Paris (2021)

PROFESSIONAL MEMBERSHIPS & AFFILIATIONS

- Member of the Canadian Mathematics Education Study Group (CMESG)
- Member of the Ontario Association for Mathematics Education (OAME)
- Student Affiliate of the Canadian Psychological Association (CPA)
 - Section(s): Education & School Psych., Teaching of Psych., Suicide Prevention, Sexual Orientation & Gender Identity, Clinical Psych., Counseling Psych., Addiction Psych.
- Student Member of the Canadian Mathematical Society (CMS)
- Student Member of the Mathematical Association of America (MAA)
 - SIGMAA(s): Math Knowledge for Teaching, Math Instruction Using the Web, Recreational Math
- Member of Spectra, the Association for LGBT Mathematicians